

Chitwan Saharia

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Education

B.Tech. Compute Science and Engineering **2015-19**
Indian Institute of Technology, Bombay, India

Professional Experience

Ideogram AI **Toronto**
Co-Founder / Researcher *December 2022 - Present*

- Working on controllable high resolution image generation.
- Focus Areas: Pre-training, Post-training, Large scale data processing, Model serving infrastructure.

Google Brain **Toronto**
Senior Research Scientist *Oct 2019 - November 2022*

- Worked on generative modeling of discrete and continuous structured data.
- Focus Areas: Image and text generation, Non-autoregressive models, Diffusion models.

Mila **Montreal**
Research Intern *May-July 2017/2018*

- Worked on benchmarking learning algorithms for Grounded Language Learning.
- Focus Areas: Imitation Learning, Goal Conditioned RL.

Publications

- Imagen Video: High Definition Video Generation with Diffusion Models**
Jonathan Ho*, William Chan*, **Chitwan Saharia***, Jay Whang*, Ruiqi Gao, Alexey Gritsenko, Diederik P. Kingma, Ben Poole, Mohammad Norouzi, David J. Fleet, Tim Salimans*
arXiv pre-print
- Synthetic data from diffusion models improves ImageNet classification**
Shekoofeh Azizi, Simon Kornblith, **Chitwan Saharia**, Mohammad Norouzi, David J Fleet
Transactions on Machine Learning Research, 2023
- Imagen editor and editbench: Advancing and evaluating text-guided image inpainting**
Su Wang*, **Chitwan Saharia***, Ceslee Montgomery*, Jordi Pont-Tuset, Shai Noy, Stefano Pellegrini, Yasumasa Onoe, Sarah Laszlo, David J. Fleet, Radu Soricut, Jason Baldridge, Mohammad Norouzi, Peter Anderson, William Chan
Computer Vision and Pattern Recognition (CVPR), 2023
- Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding**
Chitwan Saharia*, William Chan*, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S. Sara Mahdavi, Rapha Gontijo Lopes, Tim Salimans, Jonathan Ho, David J Fleet, Mohammad Norouzi*
Neural Information Processing Systems (NeurIPS), 2022 [OUTSTANDING PAPER]

- **Deblurring via Stochastic Refinement**
Jay Whang, Mauricio Delbracio, Hossein Talebi, **Chitwan Saharia**, Alexandros G. Dimakis, Peyman Milanfar
Computer Vision and Pattern Recognition (CVPR), 2022 [ORAL]
- **Palette: Image-to-Image Translation Models**
Chitwan Saharia, William Chan, Huiwen Chang, Chris A. Lee, Jonathan Ho, Tim Salimans, David J. Fleet*, Mohammad Norouzi*
SIGGRAPH Conference (2022)
Workshop on Deep Generative Models and Applications, NeurIPS, 2021
- **Cascaded Diffusion Models for High Fidelity Image Generation**
Jonathan Ho*, **Chitwan Saharia***, William Chan, David J. Fleet, Mohammad Norouzi, Tim Salimans
Journal of Machine Learning Research (JMLR), 2022
- **Image Super-Resolution via Iterative Refinement**
Chitwan Saharia, Jonathan Ho, William Chan, David J. Fleet, Mohammad Norouzi
Pattern Analysis and Machine Intelligence (PAMI), 2022
- **Non-Autoregressive Machine Translation with Latent Alignments**
Chitwan Saharia*, William Chan*, Saurabh Saxena, Mohammad Norouzi
Empirical Methods in Natural Language Processing (EMNLP), 2020
- **Imputer: Sequence Modelling via Imputation and Dynamic Programming**
William Chan, **Chitwan Saharia**, Geoffrey Hinton, Mohammad Norouzi, Navdeep Jaitly
International Conference on Machine Learning (ICML), 2020
- **Combating False Negatives in Adversarial Imitation Learning for Instruction Following**
K. Zolna*, **Chitwan Saharia***, L. Boussioux*, David H, Maxime CB, D Bahdanau, Yoshua Bengio
AAAI Conference (Student Abstract Track), 2020
NeuRIPS Deep Reinforcement Learning Workshop, 2020
- **A Tale of Two Modalities for Video Captioning**
P. Joshi*, **Chitwan Saharia***, Vishwajeet Singh, D. Gautam, G. Ramakrishnan, P. Jyothi
Workshop on Multi-modal Video Analysis & Moments in Time Challenge, ICCV, 2019
- **BabyAI: First Steps Towards Grounded Language Learning With a Human In the Loop**
Maxime CB, Dzmitry Bahdanau, Salem L, Lucas W, **Chitwan Saharia**, Thien HN, Yoshua Bengio
International Conference on Learning Representations (ICLR) 2019

Scholastic Achievements

- Secured All India Rank 148 in IIT JEE Examination (2015) among 1.5 million thousand candidates.
- Recipient of Kishore Vaigyanik Protsahan Yojna Fellowship instituted by the Department of Science and Technology, Govt. of India.